



Inheritance variation and evolution

Name: _____

Class: _____

Date: _____

Time:

Marks: **78 marks**

Comments:

1.

The following table gives the classification of four plant species.

Group	Species 1	Species 2	Species 3	Species 4
Kingdom	<i>Plantae</i>	<i>Plantae</i>	<i>Plantae</i>	<i>Plantae</i>
Phylum	<i>Spermatophyta</i>	<i>Spermatophyta</i>	<i>Spermatophyta</i>	<i>Spermatophyta</i>
Class	<i>Monocotyledonae</i>	<i>Dicotyledonae</i>	<i>Monocotyledonae</i>	<i>Dicotyledonae</i>
Order	<i>Poales</i>	<i>Fabales</i>	<i>Poales</i>	<i>Scrophulariales</i>
Family	<i>Cyperaceae</i>	<i>Fabaceae</i>	<i>Poaceae</i>	<i>Scrophulariaceae</i>
Genus	<i>Eriophorum</i>	<i>Pisum</i>	<i>Poa</i>	<i>Antirrhinum</i>
Species	<i>angustifolium</i>	<i>sativum</i>	<i>annua</i>	<i>majus</i>

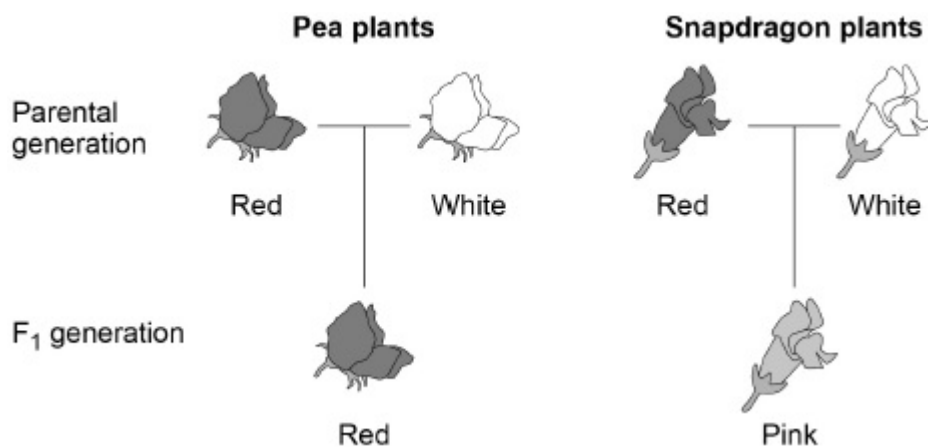
(a) Species 1 and 3 are the most closely related.

What information in the table above gives evidence for this?

(1)

Figure 1 shows the inheritance of flower colour in two species of plant.

Figure 1



- In pea plants and in snapdragon plants, flower colour is controlled by one pair of alleles.
- In **Figure 1** the parental generation plants are homozygous for flower colour.
- In heterozygous **pea** plants, the allele for red flower colour is dominant.
- In heterozygous **snapdragon** plants, the alleles for flower colour are both expressed.

Use the following symbols for alleles in your answers to parts **(b)** to **(d)**:

Pea plants

R = allele for red flowers

r = allele for white flowers

Snapdragon plants

C^R = allele for red flowers

C^W = allele for white flowers

- (b) What is the genotype of the red-flowered pea plants in the F₁ generation?

(1)

- (c) What is the genotype of a white-flowered snapdragon plant?

(1)

A gardener crossed two pink-flowered snapdragon plants.

- (d) Draw a Punnett square diagram to show why only some of the next generation plants had pink flowers.

Identify the phenotypes of all the offspring plants.

(3)

- (e) What percentage of the offspring would you expect to have pink flowers?

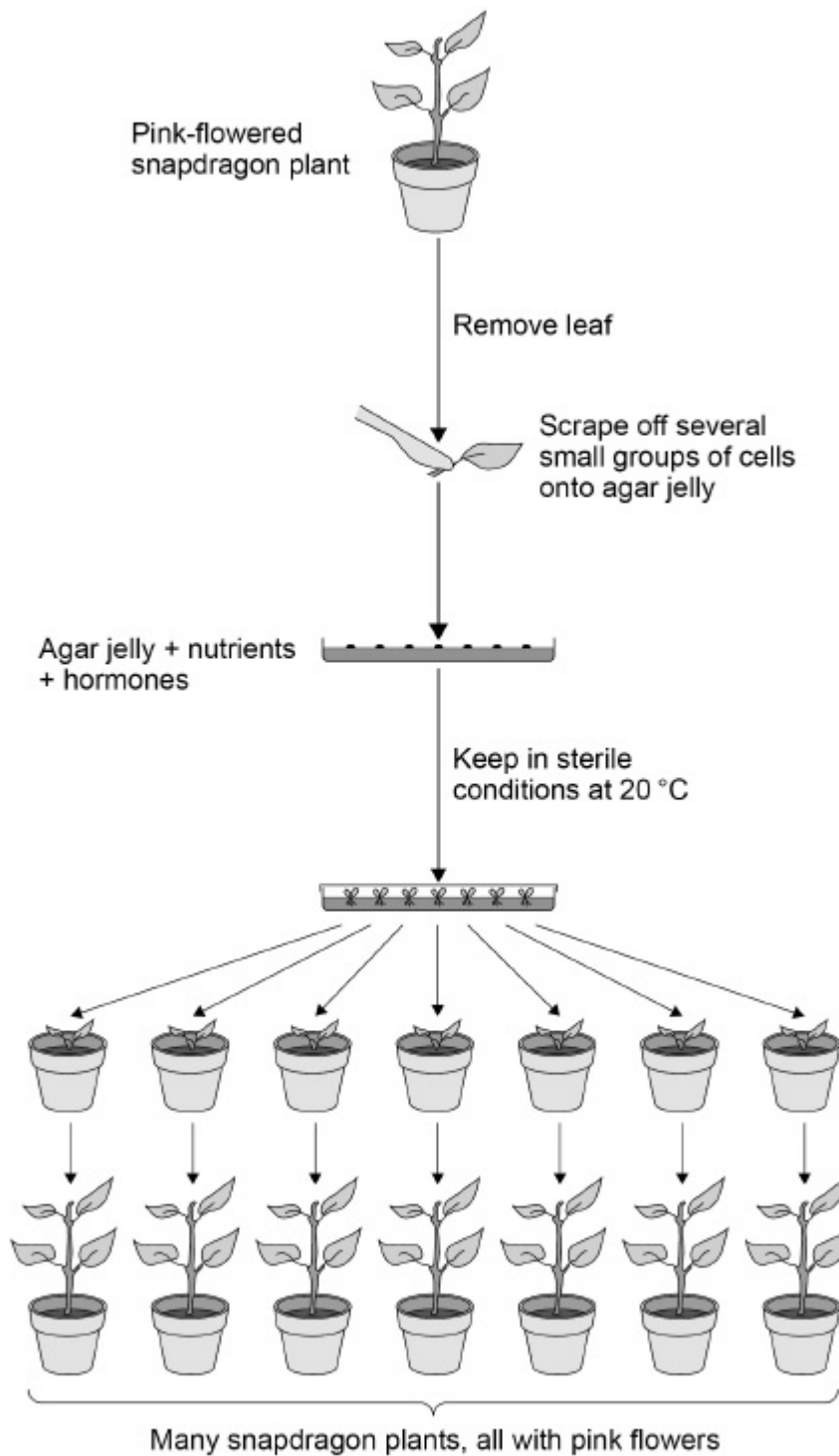
Percentage = _____%

(1)

Commercially, hundreds of pink-flowered snapdragon plants can be produced from one pink-flowered plant.

Figure 2 shows a tissue culture technique used for producing many plants from one plant.

Figure 2



- (f) Give a reason for each of the following steps shown in **Figure 2**.

Several groups of cells are scraped off the leaf:

Nutrients are added to the agar jelly: _____

Hormones are added to the agar jelly: _____

The plant cells are kept in **sterile** conditions: _____

The plant cells are kept at **20 °C**: _____

(5)

- (g) Explain why the method shown in **Figure 2** produces **only** pink-flowered plants.

(2)

(Total 14 marks)

2.

There are two types of cell division: mitosis and meiosis.

(a) Describe **three** differences between the processes of mitosis and meiosis.

1 _____

2 _____

3 _____

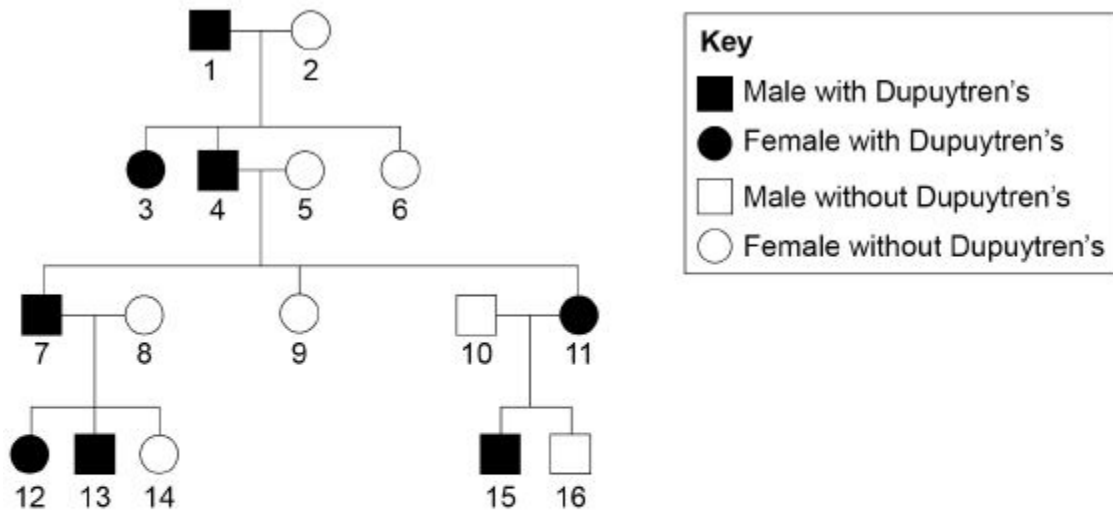
(3)

(b) Describe **one** similarity between the processes of mitosis and meiosis.

(1)

Dupuytren's is a disorder that affects the hands.

The diagram below shows the inheritance of Dupuytren's in one family.



Dupuytren's is caused by a dominant allele in this family.

D = dominant allele

d = recessive allele

(c) Give the genotype of person **1**.

Explain your answer.

Genotype _____

(2)

- (d) Person **7** and person **8** in the diagram above are expecting a fourth child.

What is the probability of the child having Dupuytren's?

You should:

- draw a Punnett square diagram
- identify which offspring have Dupuytren's

Probability = _____

(5)

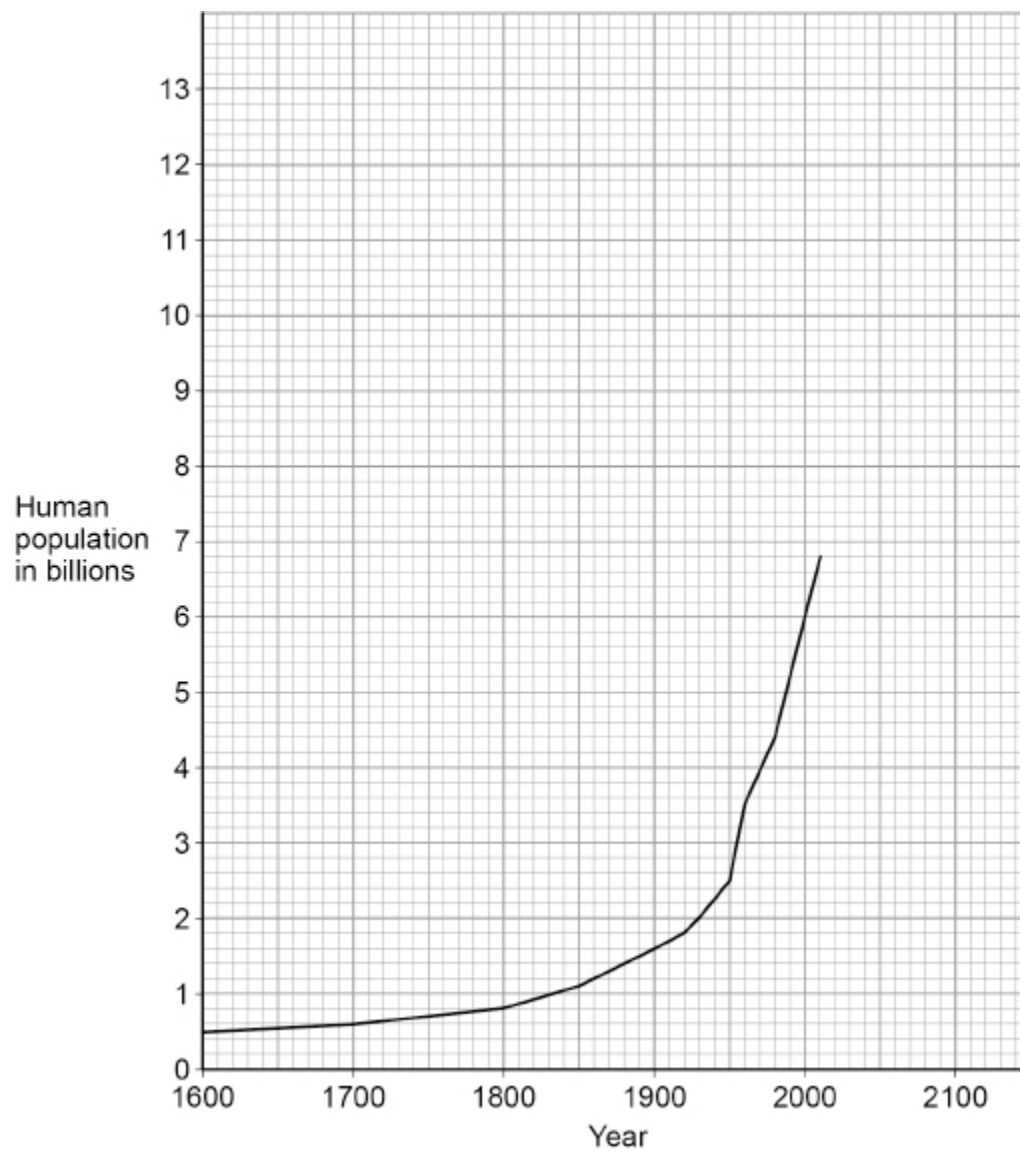
- (e) Explain how the diagram above shows the allele for Dupuytren's is **not** on the Y chromosome.

(2)

(Total 13 marks)

3.

The graph below shows the human population from 1600 to 2010.



In 1900 the human population was 1.6 billion.

- (a) Calculate how many times greater the human population was in the year 2000 compared with the year 1900.

Number of times greater = _____

(2)

- (b) In 1950 the human population was 2.5 billion.

Calculate the mean annual increase in the human population between 1900 and 1950.

Mean annual increase = _____ billion per year

(2)

- (c) Predict the human population in 2050 if the current rate of population increase continues.

You should draw an extrapolation line on the graph above.

Predicted human population = _____

(2)

- (d) The increasing human population has caused a decline in fish stocks.

Describe how fishing quotas can help to return fish stocks to a sustainable level.

(2)

- (e) Farming techniques have changed in recent years.

Describe:

- why more land is being used for farming
- how increased farming has decreased biodiversity.

(6)

- (f) Genetic modification of crop plants can help meet the demands of the increasing human population.

Golden rice is a genetically modified (GM) crop.

What is the advantage of golden rice compared with non-GM rice?

Tick (✓) **one** box.

Golden rice contains protein-rich mycoprotein

☐

Golden rice has improved nutritional value

☐

Golden rice produces human insulin

☐

(1)

- (g) Suggest **one** reason why some people are concerned about the use of golden rice.

(1)

(Total 16 marks)

4.

Phenylketonuria (PKU) is an inherited condition. PKU makes people ill.

- (a) PKU is caused by a recessive allele.

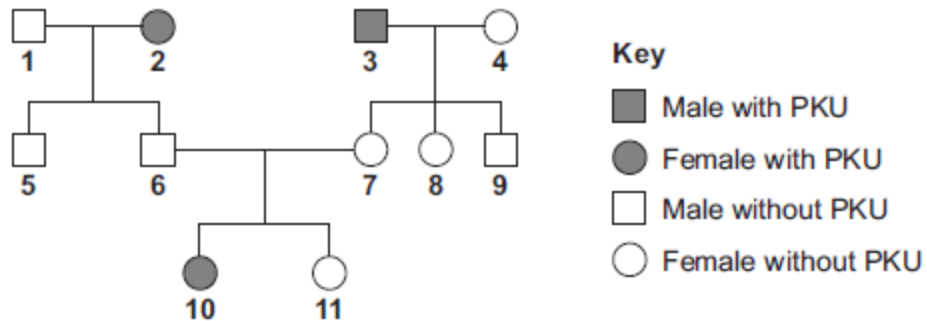
- (i) What is an allele?

(1)

- (ii) What is meant by recessive?

(1)

- (b) The diagram below shows the inheritance of PKU in one family.



- (i) Give **one** piece of evidence from the diagram that PKU is caused by a recessive allele.

(1)

- (ii) Persons **6** and **7** are planning to have another child.
Use a genetic diagram to find the probability that the new child will have PKU.

Use the following symbols in your answer:

N = the dominant allele for **not** having PKU

n = the recessive allele for PKU.

Probability = _____

(4)

- (c) Persons **6** and **7** wish to avoid having another child with PKU.

A genetic counsellor advises that they could produce several embryos by IVF treatment.

- (i) During IVF treatment, each fertilised egg cell forms an embryo by cell division.

Name this type of cell division.

(1)

- (ii) An embryo screening technique could be used to find the genotype of each embryo.

An unaffected embryo could then be placed in person **7**'s uterus.

The screening technique is carried out on a cell from an embryo after just three cell divisions of the fertilised egg.

How many cells will there be in an embryo after the fertilised egg has

divided three times?

(1)

- (iii) During embryo screening, a technician tests the genetic material of the embryo to find out which alleles are present.

The genetic material is made up of large molecules of a chemical substance.

Name this chemical substance.

(1)

- (d) Some people have ethical objections to embryo screening.

- (i) Give **one** ethical objection to embryo screening.

(1)

- (ii) Give **one** reason in favour of embryo screening.

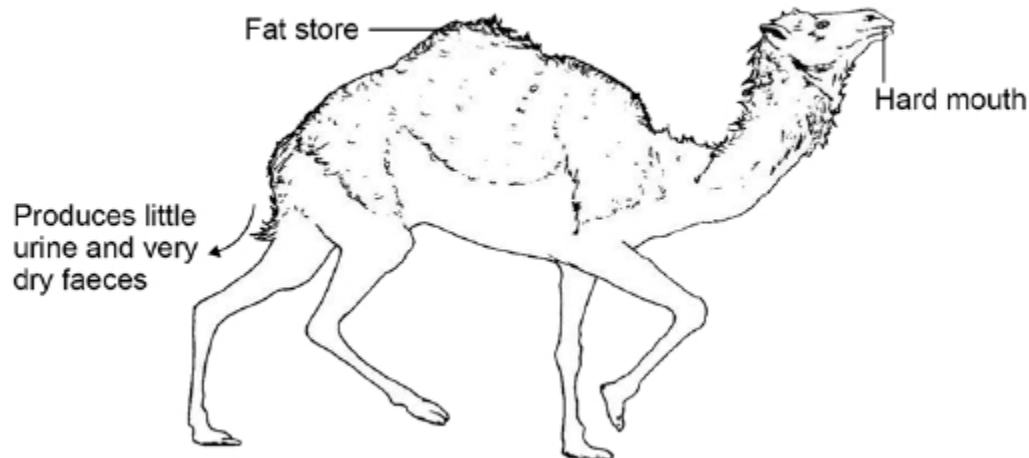
(1)
(Total 12 marks)

5.

Figure 1 shows a type of camel called a dromedary (*Camelus dromedarius*).

The dromedary lives in hot, dry deserts.

Figure 1



- (a) One adaptation of the dromedary is 'temperature tolerance'.

This means that the animal's body temperature can rise by up to 6 °C before it starts to sweat.

Explain how temperature tolerance can help the dromedary to survive in the desert.

(2)

- (b) Three more adaptations of the dromedary are given in **Figure 1**.

Give a reason why each adaptation helps the animal survive in the desert.

Fat store _____

Produces little urine and very dry faeces _____

Hard mouth _____

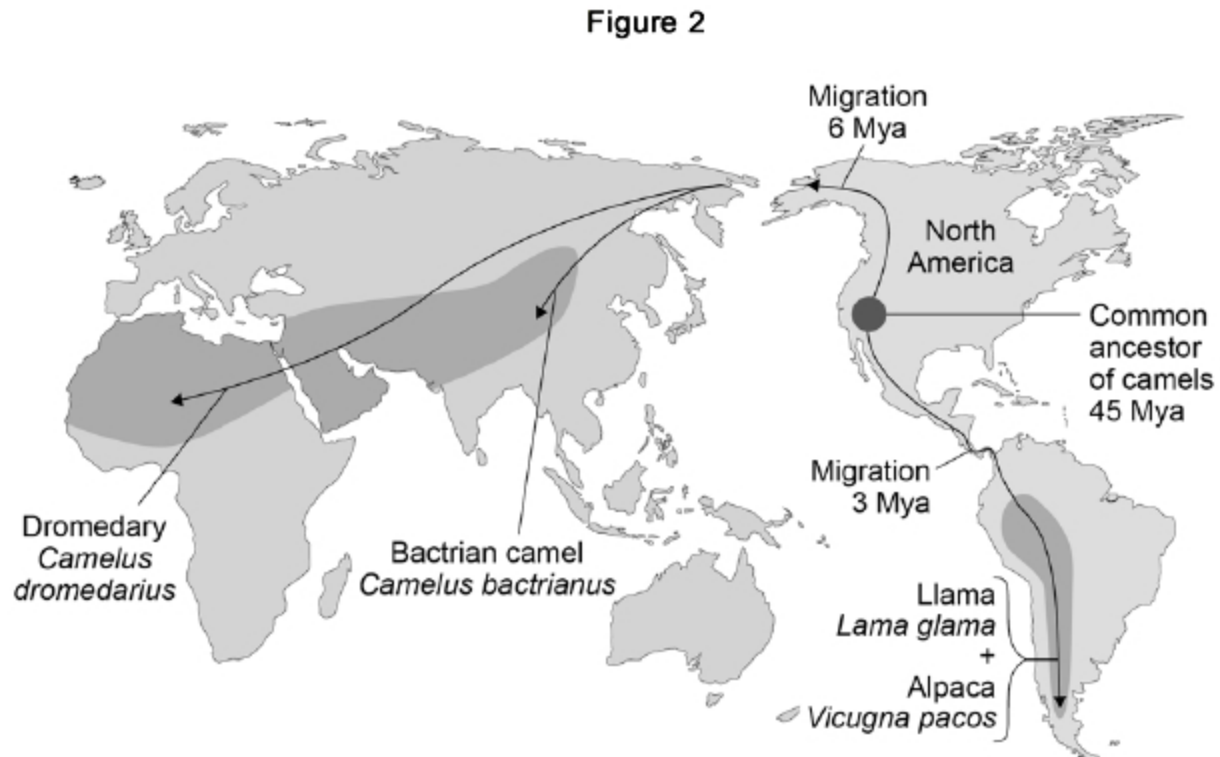
(3)

There are several species of the camel family alive today.

Scientists think these species evolved from a common ancestor that lived in North America about 45 million years ago (Mya).

Figure 2 shows:

- where four modern species of the camel family live today
- how the ancestors of these camels migrated from North America.



- (c) Which **two** of the four modern species of camel do scientists believe to be most closely related to each other?

Give the reason for your answer.

_____ and _____

Reason _____

(1)

- (d) Describe the type of evidence used for developing the theory of camel migration shown in **Figure 2**.

(2)

- (e) Explain how several different species of camel could have evolved from a common ancestor over 45 million years.

(6)

(Total 14 marks)

6.

Figure 1 shows a ring-tailed lemur.

Figure 1



The table below shows part of the classification of the ring-tailed lemur.

Classification group	Name
Kingdom	<i>Animalia</i>
Phylum	<i>Chordata</i>
	<i>Mammalia</i>
	<i>Primates</i>
	<i>Lemuroidea</i>
Genus	<i>Lemur</i>
	<i>catta</i>

(a) Complete the table above to give the names of the missing classification groups.

(2)

(b) Give the binomial name of the ring-tailed lemur.

Use information from the table above.

(1)

Lemurs are only found on the island of Madagascar.

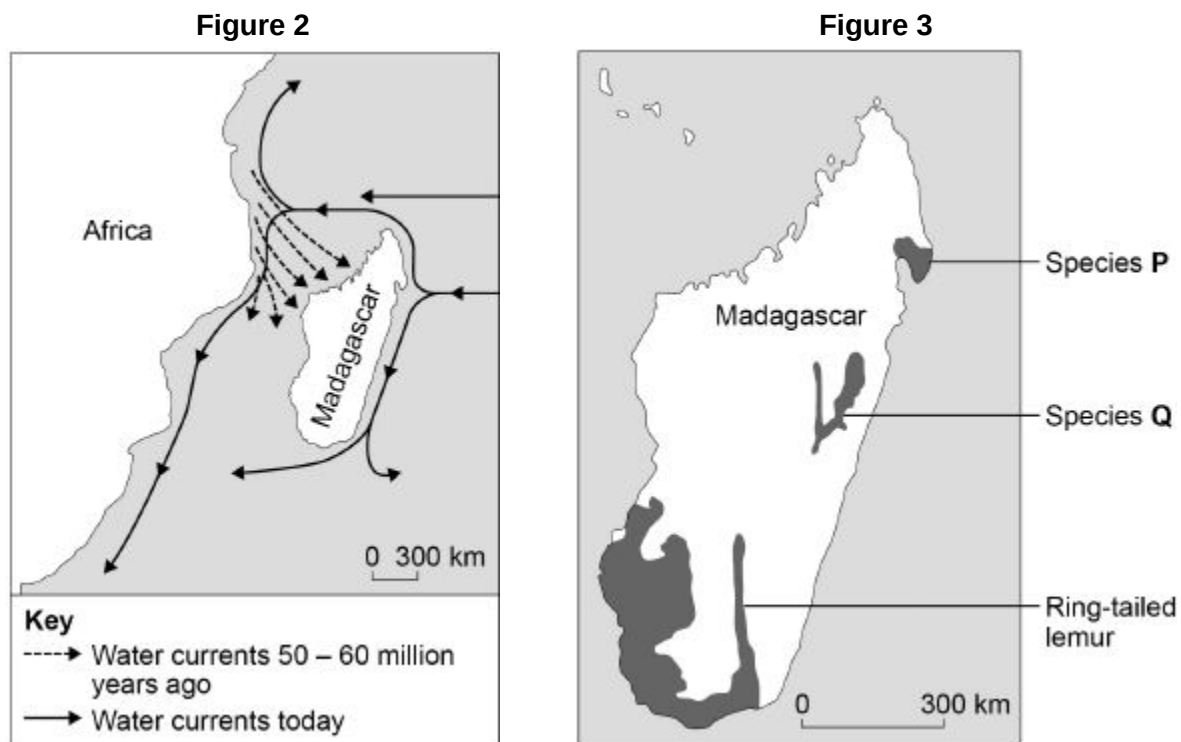
Madagascar is off the coast of Africa.

Scientists think that ancestors of modern lemurs evolved in Africa and reached Madagascar about 50-60 million years ago.

Today there are many species of lemur living on Madagascar.

Figure 2 shows information about water currents.

Figure 3 shows the distribution of three species of lemur on Madagascar.



(c) Suggest how ancestors of modern lemurs reached Madagascar.

(1)

- (d) Describe how the ancestors of modern lemurs may have evolved into the species shown in **Figure 3**.

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.

(5)

(Total 9 marks)

Mark schemes

1.

- (a) same kingdom + phylum + class + order
or
 same order
or
 they have the top four groups the same
allow both Poales

1

- (b) **Rr / rR**
do not accept RR or rr
ignore heterozygous
do not accept homozygous

1

- (c) **C^WC^W**

1

- (d)
allow R and W throughout
allow own symbols if defined

parental genotypes / gametes correct for both parents:

C^RC^W C^RC^W / C^R and C^W

1

genotypes of offspring correctly derived in a Punnett square:

C^RC^R C^RC^W C^WC^W

allow correctly derived genotypes from incorrect gametes

1

correct identification of phenotypes from their cross:

C^RC^R = red

C^RC^W = pink

C^WC^W = white

allow colours correctly identified from different offspring, only if pink and other colour(s) are given

1

- (e) answer correctly derived from part **(d)** to match stated phenotypes
allow 50(%) if no offspring given in part (d)
allow to match genotypes if no phenotypes given

1

- (f) (several groups)
 so many / several plants can be produced
allow each (group) will give a new plant

1

(nutrients)

for making protein / amino acids **or** for making chlorophyll **or for providing** energy **or for respiration**

allow other examples

*do **not** accept making energy*

ignore for growth

1

(add hormones)

so differentiation occurs **or** so roots / shoots develop

allow for the formation of different tissues / organs / named

allow to stimulate cell division

1

(sterile conditions)

to prevent growth / entry of microorganisms / named type **or** prevent decay / disease

ignore to kill microorganisms

ignore contamination unqualified

1

(temperature = 20 °C)

so optimum / good growth

allow reference to enzymes working well

ignore enzymes not denatured

ignore reference to pathogens / microorganisms

1

- (g) (all new plants have been) produced by asexual reproduction / mitosis **or** produced without (fusion of) gametes

ignore produced from one parent

1

(so) all are genetically identical / clones **or** all are **C^RC^W** / heterozygous

allow all are the same genotype / alleles / genes / DNA

1

[14]

2.

(a) any **three** from:

- mitosis produces two (daughter) cells but meiosis produces four (daughter) cells

answers must be comparative

- one cell division in mitosis but two cell divisions in meiosis
- mitosis produces cells with two of each chromosome, but meiosis produces cells with one of each chromosome

allow mitosis produces diploid cells but meiosis

produces haploid cells

*allow mitosis maintains the number of chromosomes **or***

*mass of DNA **or** mass of genetic material but meiosis*

halves the number / mass

*allow mitosis produces cells with 23 pairs **or** 46*

chromosomes but meiosis produces cells with 23

chromosomes

- mitosis produces genetically identical cells, but meiosis produced genetically different cells

allow other correct differences between the processes

of mitosis and meiosis

3

(b) any **one** from:

- DNA doubles / copies / replicates (once)

*allow chromosomes **or** genetic material **or** genetic*

information double / replicate / are copied

- increase in the number of mitochondria / ribosomes / sub-cellular structures

ignore mitochondria / ribosomes are copied / duplicated

allow chromosomes / chromatids pulled to side (of cell)

allow other correct similarities between the processes of mitosis and meiosis

1

(c) Dd / dD

allow heterozygous

1

has **D** because has Dupuytren's **and** has **d** because child / person 6 is homozygous recessive **or** does not have Dupuytren's **or** is **dd**

*allow has **D** because has Dupuytren's **and** person 1 and person 2 both passed **d** to child / person 6*

*allow has **D** because has Dupuytren's **and** cannot be homozygous / **DD** or all the children would have Dupuytren's*

1

(d) male / person 7 gametes correct: **D** and **d**

1

female / person 8 gametes correct: **d** and **d**

allow 1 mark for both sets of gametes correct if parents not identified

1

correct derivation of offspring genotypes:

Dd Dd dd dd

allow correct derivation of offspring genotypes from incorrect gametes

1

offspring with Dupuytren's identified

allow correct for genotypes stated in mp3

1

probability correct from the correct identification given

allow probability correct from offspring genotypes if identification not given

1

(e) female(s) / person(s) 3 / 11 / 12 have Dupuytren's

allow some females have Dupuytren's

1

females don't have Y chromosome

or

Dupuytren's is passed from fathers / 1 / 7 to daughters / 3 / 12, (so is not on the Y chromosome)

allow only males have Y chromosomes

allow females are XX

allow Dupuytren's is passed from mothers / 11 to children / 15, (so is not on the Y chromosome)

1

[13]

3.

(a)

$\frac{6.0}{1.6}$

allow a range of 5.9 to 6.1 for 6.0

1

3.75

*do **not** accept if a unit is given*

if no other marks awarded, allow a correct answer using a value of 5.8 or 6.2 for 1 mark

1

(b)

$$\frac{2.5 - 1.6}{50}$$

$$\begin{array}{l} \text{allow} \\ 0.9 \\ \hline 50 \end{array}$$

1

0.018 (billion per year)

1

(c) suitable extrapolation line drawn on the graph.

allow straight extrapolation

1

reading taken at 2050 from student's line

allow a tolerance of $\pm \frac{1}{2}$ small square

allow 1 mark for 10 billion if no extrapolation drawn

1

(d) fewer fish caught **or** limit the number of fish caught

*allow a method of doing this, eg increase mesh size **or***

do not catch young fish

1

(remaining fish) can reproduce

allow more fish (survive to) reproduce

1

(e) **Level 2:** Scientifically relevant facts, events or processes are identified and given in detail to form an accurate account.

4–6

Level 1: Facts, events or processes are identified and simply stated but their relevance is not clear.

1–3

No relevant content

0

Indicative content

human land use

- increasing population requires more food
- crops / livestock for food
- farming crops for biofuels
- peat use as compost
- peat use as fuel
- increased use of pesticide / insecticide / herbicide / fertilisers
- use of free-range / organic methods increases land use (for same yield)

link to biodiversity

- deforestation
- monocultures
- loss of hedgerows to make fields larger
- loss of habitat
- consequence of loss of habitat e.g. (change in) migration
- fertiliser run off polluting water
- use of pesticide / insecticide / herbicide reduces insects / plants which damages food chains
- more soil erosion

link to atmospheric pollution

- more carbon dioxide (from farm animals / machinery)
- more methane (from cows)
- climate change **or** global warming
- example of impact on biodiversity
- acid rain
- desertification

Answers referring to only land use or only biodiversity are level 1

(f) golden rice has improved nutritional value

1

(g) any **one** from:

- gene may contaminate / enter other breeds / species
- reduction / extinction of population of wild / traditional rice
- reduction / extinction of population of flowers / insects
- high cost of seeds
- *allow decrease in biodiversity*
- may have too much vitamin A (in diet)
 - allow decrease in gene pool*
 - allow may harm (human) health*
 - allow may cause side effects (on humans)*
 - ignore references to religious beliefs*
 - ignore may harm humans unqualified*

1

[16]

4.

- (a) (i) one form of a / one gene
- do **not** allow 'a type of gene'*
- allow a mutation of a gene*

1

(ii) not expressed if dominant / other allele is present / if heterozygous

or

only expressed if dominant allele not present / or no other allele present

allow need two copies to be expressed / not expressed if only one copy / only expressed if homozygous

1

(b) (i) two parents without PKU produce a child with PKU / **6** and **7** → **10**

allow 'it skips a generation'

1

(ii) genetic diagram including:

accept alternative symbols if defined

Parental gametes:

6: **N** and **n**

and 7: **N** and **n**

1

derivation of offspring genotypes:

NN Nn Nn nn

allow genotypes correctly derived from student's parental gametes

1

identification: **NN** and **Nn** as non-PKU

OR nn as PKU

allow correct identification of student's offspring genotypes

1

correct probability only: 0.25 / $\frac{1}{4}$ / 1 in 4 / 25% / 1 : 3

*do **not** allow 3 : 1 / 1 : 4*

*do **not** allow if extra incorrect probabilities given*

1

(c) (i) mitosis

correct spelling only

1

(ii) 8

1

(iii) DNA

allow deoxyribonucleic acid

*do **not** allow RNA / ribonucleic acid*

1

(d) (i) may lead to damage to embryo / may destroy embryos / embryo cannot give consent

allow avoid abortion

allow emotive terms – eg murder religious argument must be qualified

allow ref to miscarriage

allow idea of avoiding prejudice against disabled people

allow idea of not producing designer babies

1

(ii) any **one** from:

- prevent having child with the disorder / prevent future suffering / reduce incidence of the disease

ignore ref to having a healthy child

ignore ref to selection of gender

- embryo cells could be used in stem cell treatment

allow ref to long term cost of treating a child (with a disorder)

allow ref to time for parents to become prepared

1

[12]

5.

(a) less sweating so less water loss

1

(as) no / little water available in desert

1

(b) (fat store) can be metabolised / respired to water

1

(little urine...) conserve water

1

(hard mouth) not damaged by spines on plants / on food

or

not damaged by hard / dry food

1

(c) dromedary / *C.dromedarius*
and bactrian / *C. bactrianus*

no mark for the names, but must be identified

because

same genus

ignore 'both are Camelus'

1

(d) any **two** from:

- the fossil record
- oldest fossils in N. America
- or
- newer fossils in S. America / in Asia / in Africa
*allow numbers for ages (45 Mya **and** 3 Mya / 6 Mya)*
- chemical / DNA analysis of living species
allow radioactive dating of fossils

2

(e) isolation of separate camel populations by sea
or
by mountains

1

habitat variation / described between populations

allow examples – biotic (e.g. food / predators) or abiotic

1

genetic variation / mutation in each population

1

45 million years is sufficient time to accumulate enough mutations

1

natural selection

or

better adapted survive to reproduce

1

pass on favourable allele(s)

allow gene(s)

1

[14]

6.

(a)

Classification group	Name
Class	<i>Mammalia</i>
Order	<i>Primates</i>
Family	<i>Lemuroidea</i>
Species	<i>catta</i>

all 4 correct = 2 marks

2 or 3 correct = 1 mark

0 or 1 correct = 0 marks

2

(b)	Lemur catta	
	<i>ignore capitalisation / non-capitalisation of initial letters</i>	
	<i>ignore italics / non-italics</i>	
	<i>ignore underlining / non-underlining</i>	1
(c)	carried by (favourable) currents on masses of vegetation	
	<i>allow description of currents from Figure 2</i>	
	<i>ignore swimming</i>	1
(d)	isolation of different populations	1
	habitat variation between lemur populations	
	<i>allow examples – biotic (e.g. food / predators) or abiotic (e.g. temperature)</i>	1
	genetic variation or mutation (in each population)	1
	better adapted survive (reproduce) and pass on (favourable) allele(s) to offspring	
	<i>allow natural selection or survival of the fittest and pass on (favourable) allele(s) to offspring</i>	
	<i>allow gene(s) / mutation as an alternative to allele(s)</i>	1
	(eventually) cannot produce fertile offspring with other populations	
	<i>allow cannot reproduce 'successfully' with other populations</i>	
	<i>ignore cannot reproduce unqualified</i>	1
		[9]

Examiner reports

1.

- (a) Given a table showing the complete classification of four flowering plants, most students had no difficulty on deciding that species 1 and 3 were the most closely related because they belonged to the same order (the other two only sharing taxonomic groupings down to class level).
- (b)(c) Figure 1 showed how a red-flowered pea plant crossed with a white-flowered pea plant produced offspring (F_1) plants which all had red flowers, whereas a similar cross between red-flowered and white-flowered snapdragon plants produced offspring with only pink flowers.

Using the given symbols for alleles, students then had to give the genotype of an F_1 red-flowered pea plant and of a white-flowered snapdragon plant. 74% of students successfully gave Rr for the pea plant, and 54% managed C^WC^W for the snapdragon plant.

Although the format of the given allele symbols C^R and C^W may have appeared unusual to most students in this application question (AO2), all they had to do was select and copy them. Errors included ' C^W ', ' C^R ', ' C^W ', ' r ', ' C^{WW} ' and ' C^{RW} '.

- (d) Students had to draw a Punnett square diagram to show how a cross between pink-flowered snapdragon plants produced only *some* plants with pink flowers in the next generation. Around 46% of students were completely successful. Many did not gain the mark for assigning phenotypes to the offspring genotypes as they did not do so, despite the instruction in the question to do so.

A few students decided to use their own symbols for alleles rather than the ones given in the question – this would have been legitimate and creditworthy if they had first been defined, but generally they were not and thus a mark could not be gained.

- (e) The percentage of offspring with pink flowers given in answer to this question had to match that predicted by the student's answer to part (d). Had the answer to (d) been completely correct, then this would have been 50%. About 83% of students got this right.
- (f) This question examined practical techniques involved in the cloning of pink-flowered snapdragon plants, as outlined in Figure 2.

Most students were unable to give reasons for the first three techniques:

- why several groups of cells were used (i.e. to produce several new plants)
- why nutrients were added to the agar jelly (i.e. for making a named organic substance – e.g. protein or chlorophyll – or to provide energy for growth)
- why hormones were added to the jelly (i.e. to stimulate differentiation or the growth of roots or shoots). Most students thought 'to encourage growth' was sufficient.

Many students, however, understood that sterile conditions were necessary to prevent the growth of microorganisms or to prevent disease, and that a temperature of 20°C would encourage optimum or good growth or provide a temperature at which enzymes worked well. The 'prevention of growth of pathogens' was an inappropriate concept here as, unlike a human, a plant does not have a body temperature of 37°C.

- (g) To explain why the technique shown in Figure 2 produced only pink-flowered plants, many students stated correctly that the offspring were 'clones' or that all would have been 'genetically identical'. The idea that asexual reproduction or cells produced by mitosis was involved was given far less often. 81% of students scored at least one mark.

2.

This question was about cell division and genetics. In (a) many students confused the two types of cell division, mitosis and meiosis: 1 in 10 students were able to give 3 correct differences between the two. Some described only one type of division rather than giving the comparison demanded by the question; others fell short of giving sufficient detail – e.g. describing the daughter cells as 'identical / non-identical' rather than being *genetically* identical / non-identical; many included irrelevant points about where in the body the cell division occurred (the question asked for differences in the 'processes'). Two common misunderstandings were that meiosis occurred 'in gametes' and that meiosis required two parents whereas mitosis required only one.

For (c) explanations of how students derived the genotype of person 1 were frequently insufficient. Since this person had the condition Dupuytren's, he must have had either one or two of the dominant alleles. The fact that he had an unaffected child (person 6), showed he must also have had the recessive allele – hence his genotype was **Dd** or heterozygous. Most students mentioned only the children as evidence and did not relate person 1's genotype to his phenotype.

Students had to draw a Punnett square diagram in (d) to find the probability that persons 7 and 8 in could produce another child with Dupuytren's. Since persons 7 and 8 had *different* phenotypes, it was surprising how many students assigned the *same* genotype to each of them (usually both being **Dd**). Other common errors were not identifying which parent was which (i.e. gametes **D** and **d** from person 7, and only **d** gametes from person 8) and not assigning phenotypes to the offspring genotypes.

In (e) explanations of the evidence in the pedigree diagram that the **D** allele is *not* on the Y chromosome, it was evident that many students did not know that males are XY and females XX as many thought the female had a Y chromosome. Those that did know this were able to point out that some females had Dupuytren's despite not having a Y chromosome, or that the condition can be passed from fathers to their daughters, or that an affected mother and unaffected father can have an affected son. Many students insisted, erroneously, that the allele must have been on the X chromosome.

3.

The question was about human population increase and the problem of provision of sufficient food. For the calculation in (b), both of the values required were given in the information, so it was not necessary to determine figures from the graph. Despite this, many students did attempt to, often incorrectly. For those who did show the correct numbers in their calculations, a large proportion used them incorrectly, often adding them and very few did the necessary calculation of dividing the difference by 50. Some just left the value at '0.9' and a significant number divided this by 5, rather than by 50. Others calculated the mean of the two given values rather than the mean annual increase.

In (c) most students attempted to draw some sort of extrapolation on the graph; however, this was often not linked to the trend shown by the line, rising a little, then plateauing or even taking a downturn. Students could still gain the second mark for reading the correct value from their own line. Often this was not done accurately, particularly by students who continued their extrapolation beyond 2050 and read off the final population value. It was clear that students who simply read the scale, and gave answers such as '10' had not considered the implication that their answer of a human population of only 10, rather than 10 billion, is unlikely.

Of the more successful students in (d) these showed some understanding of the term 'quota' and described limiting fishing in one way or another. However, this was as far as some got as they omitted the idea that the remaining fish would be able to reproduce, in order to maintain stocks. Lower-attaining students often suggested that a quota referred to the amount of fish that people could buy or eat. In (e) many students did not link the need for more farmland to an increasing population needing more food. References to the use of chemicals to enhance growth were only given very occasionally. In terms of reduced biodiversity, hardly any students went beyond 'cutting down trees' and habitat loss. Few students referred to methane production but these rarely went any further. Whilst many students filled up the available space, most of these simply repeated the same vague points in order to do so.

4.

- (a) (i) In many cases, it was not possible to tell whether students understood the concept of an allele, i.e. that it was one alternative form of a particular gene. This was due largely to imprecise communication by the student. Thus, less than one-third scored the mark.
- (ii) There was slightly more success in explaining what was meant by recessive, but even here there was confusion between an allele being 'inherited' and it being 'expressed', it not being correct to say that the allele would not be 'inherited' if the dominant allele was present. More able students made their ideas clearer by explaining that two copies would need to be inherited before the condition determined by the allele was expressed, or that it would only be expressed if the individual was homozygous, or if not heterozygous.
- (b) (i) Students had to find one piece of evidence from the diagram which showed that PKU was caused by a recessive allele. Most concentrated on the fact that a sufferer was shown to have three children without PKU; however, this would also have been possible had the allele been dominant and the sufferer heterozygous. The only evidence for the allele being recessive was that two parents without the condition produced a child with it. Some students gave an alternative satisfactory version of this by stating that PKU 'skipped a generation'. Just over one-third of students got this right.
- (ii) Almost half the students scored full marks here, with a further one-third scoring 3 out of the 4 marks available. The most common omission was a failure to identify which of the offspring genotypes would result in the phenotype of having PKU. Occasionally, students gave more than one version for the probability of the offspring being a sufferer which was fine if all of the answers were correct, but should be discouraged as examiners will not choose between a correct and an alternative incorrect answer.
- (c) (i) A little over a third of students knew that the fertilised egg cell divided by mitosis.
- (ii) About the same proportion could work out that, after 3 rounds of cell division, 8 cells would have been produced.
- (iii) It was rather disappointing to find that only half the students could give DNA as the chemical substance that made up the genetic material.
- (d) (i) Suggested ethical objections to embryo screening generally centred around the termination of a human life or the fact that this new life form was not in a position to give consent to the testing. Answers which referred to 'playing God' or to not being 'natural' were not given credit unless they were further qualified. Over two-thirds of students were able to make a sensible suggestion.
- (ii) Suggested reasons in favour of embryo screening included; reference to preventing future suffering by a child with the condition, cost-saving by the Health Service, and the potential use of cells from unused healthy embryos in stem cell treatment. A little over half of the students gave a satisfactory answer. A common incorrect idea was that detection of the condition meant that it could then be cured in that individual embryo.

6.

- (a) 37% of students knew that the four missing taxa in the hierarchy were class, order, family and species. Some demonstrated the use of mnemonics to enable them to recall these technical terms. Unfortunately, some could only remember the mnemonic, not what it represented.
- (b) 48% of students knew the binomial name of an organism was composed of the genus plus species: here, *Lemur catta*. Many incorrect answers included the family name from the table, '*Lemuroidea*'. Many gave a single word for the binomial name.
- (c) Most students appeared not to make use of the scale given on the map. Therefore, did not realise that the ancestral lemur would have had to swim 400 kilometres to get to Madagascar.

Others did not seem to appreciate how long ago 60 million years was and stated that the ancestral lemurs might have been stowaways on boats. 3% of students described how lemurs would have likely been carried on rafts of vegetation (or tree trunks) which would have floated and been propelled by the favourable water currents.

- (d) The ideas required to answer this question were that several populations of ancestral lemurs would have been isolated from one another, perhaps by mountains, deserts or rivers, and found themselves in different types of habitat. Genetic variation, or mutation, within each population then led to the better adapted surviving long enough to reproduce and pass on their beneficial alleles to their offspring, eventually producing different groups of lemurs that could not reproduce with members of another group to produce fertile offspring. Such a complete explanation was given by 4% of students. Although 53% of students were able to achieve at least two of the five available marks, usually for the first two marking points given above.

Weaker explanations gave ideas such as 'the lemurs were in different parts of Madagascar', 'the lemurs survived and passed on their characteristics', 'the lemurs could no longer reproduce together'.